Shi Luode

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Research Interests: Neural Combinatorial Optimization, Reinforcement Learning, Large Language Models, and Robotics

EDUCATION

National University of Singapore (NUS) ***08/2025–Present***

Master of Science in Robotics | GPA: 4.07/5.00 | Expected graduation: Nov 28, 2026

- Research focus: autonomous exploration, reinforcement learning, decision optimization, viewpoint planning, and graph-based policy learning.
- Relevant coursework: Topics in Robotics (A+), AI Planning and Decision Making, Cognitive Robotics, Robot Dynamics and Control, and Machine Learning in Robotics.

Shandong University (SDU) ***09/2021–06/2025***

B.Eng. in Optical Information Science and Engineering | GPA: 3.61/5.00 | Rank: 7/43

RESEARCH EXPERIENCE

Preference-Adaptive Multi-Agent Optimization for Multi-Objective Combinatorial Optimization ***04/2026–Present***

Manuscript under review, MARMoT Lab, National University of Singapore

Authors: **Luode Shi***, Conghui Wang, Mingfeng Fan, and Guillaume Adrien Sartoretti*

- Research problem: a shared preference sets the desired balance between objectives, such as shorter total travel versus more balanced routes, but each vehicle or machine has a different role. Developed PAMAO to adapt decisions to each agent and to learn from good solutions found under nearby trade-offs.
- Method design: integrated Agent-wise Preference Injection (AWPI) and Collaborative Self-Improvement (CSI) into the MACSIM and PARCO-WECA backbones without replacing their encoders or action decoders, forming PAMAO-M and PAMAO-P, respectively.
- Experimental results: on generated routing benchmarks, PAMAO-M (MACSIM + PAMAO) and PAMAO-P (PARCO-WECA + PAMAO) each improved solution quality over their respective base backbones. On generated scheduling tests, PAMAO-M also improved over MACSIM. Under DCAN's official greedy protocol, PAMAO-D (DCAN + PAMAO) improved over the DCAN backbone in most tested two- and three-objective settings, including larger scales.
- Public benchmarks: on public routing and scheduling benchmarks, PAMAO-M improved average solution-set quality over the MACSIM-WECA base.

Ariadne: Reinforcement Learning for Single-Robot Autonomous Exploration ***12/2025–05/2026***

Research Project, MARMoT Laboratory, National University of Singapore

Advisor: Asst. Prof. Guillaume Sartoretti | Robotics, Autonomous Exploration, Reinforcement Learning

- Goal: enable a robot to explore an unfamiliar map by choosing useful observation points, rather than repeatedly moving to the nearest unexplored boundary.
- Planning system: generated candidate points around map boundaries, snapped them to traversable locations, removed duplicates, ranked them by newly visible area, and grouped nearby candidates; capped the action list at 25 points per decision.
- Training method: built an expert planner that selects points with low overlap in newly visible regions and orders them to reduce travel. Combined its routes with exploration-coverage and travel-distance rewards to train a Soft Actor-Critic policy with two value networks.
- Implementation and evaluation: integrated the map graph, candidate planner, policy, and parallel simulation in Ray. Tested on 100 maps with up to 600 decisions per run and recorded map-completion rate and travel distance.

ADDITIONAL EXPERIENCE

Machine Vision Intern, Shandong Institutes of Industrial Technology ***07/2024–08/2024***

Industrial inspection and machine vision

- Built a production-line inspection workflow for open glass bottles, from camera calibration and image threshold setup through defect filtering and result reporting.
- Used MXVision and image-region size/shape rules to identify empty positions, inverted bottles, and bottles that had not been opened; tuned the checks with production samples.
- Processed inspection results with C# scripts and delivered bottle counts to the client; also assisted with defect-inspection tasks for Hikrobot products.

TECHNICAL SKILLS

- **Reinforcement Learning:** decision-process modeling, Soft Actor-Critic, twin value networks, reward design, expert-route imitation, and policies guided by objective trade-offs.
- **Multi-Agent Optimization:** multi-agent routing and job-shop scheduling; preference-guided neural solvers, agent-role adaptation, collaborative training, and solution-quality evaluation.
- **Robotics and Exploration:** occupancy-map exploration, frontier and viewpoint selection, graph-based planning, candidate-action generation, route coverage, and A*-based path planning.
- **Machine Learning Systems:** Python, PyTorch, Transformer models, Ray, Hydra, TensorBoard, Lightning/RL4CO, parallel simulation and training, and unit testing.
- **Development Tools:** AI-assisted coding tools for code reading, experiment scripting, and debugging.
- **Languages:** IELTS 6.5; CET-4 and CET-6.

HONORS AND AWARDS

Optoelectronics Alumni Scholarship, School of Information Science and Engineering, Shandong University ***2025***
Second-Class Academic Scholarship, Shandong University ***2024***
Outstanding Participant, SIOM Summer Camp, University of Chinese Academy of Sciences ***2024***
Second-Class Academic Scholarship, Shandong University ***2023***
Third-Class Academic Scholarship, Shandong University ***2022***